



PRIVACY INFRASTRUCTURE FOR SOLANA.
RECURRING PAYMENTS BUILT IN.

v1.0.2 • TX-OPACITY RELEASE
Volta Team • Slashy Fx • Solo, full-time
Dogpatch • Superteam • June 2026

Crypto rails leak. Card rails take 12%.

A merchant accepting subscriptions today picks one of two losing trades:

Public chains

Solana Pay, Coinbase Commerce, every standard L1 rail.

Every customer's wallet history is on display.

The merchant sees it.

The chain sees it.

Anyone querying the chain sees it.

Card processors

Stripe, PayPal, Patreon, Apple / Google in-app.

2.9% to 12% rake.

The processor surveils every transaction.

Chargebacks are a loaded gun the buyer holds.

No recurring privacy rail

Mixers (Tornado), shielded pools (Aztec, Zcash).

Privacy primitives exist but no merchant SDK, no recurring billing, no Solana speed. Built for transfers, not subscriptions.

Three lines crossed in 18 months.

2024 Q3

Compute budget bump

Solana raised the per-tx CU ceiling.
On-chain STARK verification became economical (~900K CU, <0.001 SOL per verify).
Custom FRI verifiers in 801 KB SBF binaries became feasible.

2024 AUG

ML-KEM-768 finalised

NIST published the post-quantum KEM standard.
Hybrid X25519 + ML-KEM-768 stealth addresses now build on a standard KEM.
ML-KEM resists Shor, but the web app re-derives its keys from one Ed25519 wallet signature: they are only as safe as that wallet key. Spend authority is Ed25519 too.

2024-2026

Creator-rail revolt

Patreon's 12% rake + Apple's 30% in-app + Stripe 2.9%.
Creators publicly leaving for crypto rails (Pump.fun, Hivemapper, indie SaaS).
The market is asking for an exit.

The cryptography is ready. The chain is ready. The market is ready.

Private recurring crypto subscriptions.

Drop-in SDK for any online merchant.

Customer subscribes from the Protocol 01 wallet (or via QR scan).

Merchant receives SOL or USDC at their own retailer wallet.

Between the two: a denominated pool, STARK proofs checked on chain, hybrid post-quantum stealth addresses.

Today a subscription stays linkable to the customer's wallet: the wallet funds the one-time key in the clear (README, "The pool hides neither today").

0.80%

TOTAL ROUND-TRIP FEE

0%

MERCHANT PLATFORM CUT

23.0 s

SUBSCRIBE, MEASURED ON
DEVNET

PQ

PROOFS & KEM; KEYS STILL
FROM ED25519

Three integration paths

P01-JS

Drop-in pay button + React widgets. 5 lines
of code on the merchant's site.

MERCHANT-SDK

Server-side: register service, poll payments,
watch vaults, issue access tokens.

AUTH-SDK

"Login with Protocol 01".
QR + biometric, gate by active subscription.

Functional prototype, on-chain. Not yet adopted.

The rail is real and end-to-end.

No external merchant has integrated yet, and that is the honest stage we are at.

The cryptography and the SDK are ahead of the adoption curve.

What is proven, today

- 4 Anchor programs on the devnet product path · 8 STARK AIRs in a custom 801 KB FRI verifier (BENCHMARK-2026-09-13)
- Registry program live: QaQwpvBi1EQpevNE21D2oNBHFsLtoLwa7aXH26zRhQB
- 22/22 live-devnet E2E checks: register → fetch → deregister, every field byte-equal, rent refunded
- 7 packages on npm with their source in the repo · 26/26 fresh-project tarball install · per-suite test counts in HANDOFF-2026-09-13 §4c
- Mobile (Android), Chrome extension, web — all three clients shipped

```
// real on-chain signatures (devnet)
```

```
register 3odKFeJXyvpVcy7q6Df5tzqM627DE6gt5...  
deregister 4MPXSxxRyocawcGuYg93aWUVKj9WPvYx2...
```

What is NOT proven, yet

- **External merchant adoption.**
No real online business has integrated Protocol 01 today.
- The five entries seeded on the devnet registry (netflix-standard, disney-plus, etc.) are **demo placeholders** we created ourselves to validate the registry mechanism.
Not partnerships.
- That crypto-native recurring privacy is a market need **today**.
Web3 has not reached the threshold where every merchant must adopt it.

The bet.

Build the rail before the wave.
When privacy + post-quantum + creator-fee fatigue cross, the merchants who need it will find an SDK that has been hardened for 12 months.
Not one being written that day.

Five lines on the merchant's server.

Frontend ships a drop-in widget.
Backend runs a 30-second poll loop.

Frontend · React

```
import { P01Provider, SubscriptionWidget }  
  from '@protocol-01/p01-js/react';  
  
<P01Provider config={{ merchantId: 'acme' }}>  
  <SubscriptionWidget tiers={[  
    { id: 'pro', price: 19.99,  
      interval: 'monthly' }  
  ]} />  
</P01Provider>
```

Backend · Node / Cloudflare

```
import { pollPaymentsForRetailer, issueAccessToken }  
  from '@protocol-01/merchant-sdk';  
  
setInterval(async () => {  
  const r = await pollPaymentsForRetailer(c, retailer);  
  r.forEach(p => grantAccess(p));  
}, 30_000);
```

What the merchant can find today (v1, devnet)

LINKABLE The customer's wallet: absent from the subscription transaction, but it funds the one-time key in the clear, three RPC calls away

LINKABLE The note spent: matchable to the deposit that published its commitment

PUBLIC The subscription vaults: anyone can enumerate a merchant's subscribers

4 programs on the product path. 1 custom STARK verifier.

Our deployed programs

VERIFIER	p01_stark_verifier – FRI, 8 AIRs
CORE	zk_shielded – pool + sub vault
CORE	p01_registry – merchant directory
RAILS	p01_relayer – fee-paying relay
GONE	fee-splitter closed 2026-09-13; six unused crates (arcium, liquidity, zkspl, stream, subscription, whitelist) removed 2026-09-23

Solana primitives leveraged

Anchor 0.32	DSL/framework, all 4 program crates
SPL Token	USDC + native SOL handling
PDAs	deterministic registry / vault keys
Compute Budget	1.4M CU for STARK verify
SystemProgram	native SOL + account allocation
Memo Program v2	invoice memos for one-shots
Helius RPC	getProgramAccounts pagination
Privy	embedded wallet onboarding

The custom FRI/STARK verifier fits in an **801 KB** SBF binary. 890,643 CU for a circuit 7 verification, both phases in one transaction (BENCHMARK-2026-09-13 §6).

Built today, in under five months, alone.

4PROGRAMS ON THE
PRODUCT PATH**7**PACKAGES ON NPM, SOURCE
IN REPO**8**

STARK AIRS

18.6 sPOOL DEPOSIT (SHIELD),
DEVNET

Shipped

- 4 Anchor programs on the devnet product path, each deployed at its declared address
- Custom FRI/STARK verifier (801 KB SBF, 8 circuits)
- Mobile app v1.0.3 (Android APK, GitHub releases)
- Chrome extension MV3 with on-device WASM prover
- Relayer-submitted spends, scrubbed events, uniform proofs (v1.0)
- Web marketing site · design doc
- 5 verified merchants on devnet registry

Next milestones

- External audit cycle (OtterSec / Neodyme — quote in flight)
- Mainnet deployment + first 5 paying merchants
- iOS TestFlight beta
- Confidential amounts on the denominated pools

Biggest risk

Audit timing.
Privacy primitives without an audit are theatre.
We do not deploy to mainnet without one full external cycle complete.

0.80% round-trip. Zero platform cut.

0.30%

SHIELD FEE

0.50%

UNSHIELD FEE

0.80%

ROUND-TRIP TOTAL

Versus the rails merchants use today

PROVIDER	TAKE RATE	PRIVACY	SETTLEMENT
PROTOCOL 01	0.80%	POOL, LINKABLE TODAY	18.6 s / 23.0 s / 20.8 s (deposit / subscribe / withdraw, devnet)
Stripe (cards)	2.90%	NONE	2-7 days
Patreon	8-12%	NONE	Monthly
Coinbase Commerce	1.00%	NONE	~30 min finality
BTCPay self-host	0.00%	PSEUDONYMOUS	10-60 min
Solana Pay (raw)	0.00%	NONE	~1s

Worked example. SaaS at €19.99/mo, 1,000 subs, 12 months, 10% churn. Gross €143,449 → net to merchant **€142,668** on Protocol 01. Same revenue costs ~€5,460 on Stripe, €14k-17k on Patreon.

Built by hand.

Cryptography I implemented from first principles.
Not glued together from libraries:

- Custom on-chain **FRI/STARK verifier** (Goldilocks field, Poseidon AIR, 8 circuits, 801 KB SBF binary)
- Hybrid **X25519 + ML-KEM-768** stealth address construction (own design, not a reference port)
- Blinding masks on all 8 circuits: every committed proof value measured uniform by the X5 test; the simulation argument is being redone, since the next-row openings it treated as hidden are published (docs/zk-simulation-argument.md)
- Deterministic **HKDF note-secret derivation** — one seed, every shielded note recoverable cross-device
- **4 program crates**, 7 packages on npm with their source in the repo, three clients — all written by the same person

***"Financial privacy is a right, not a feature.
No team, no funding, just execution."***

— Slashy, founder

The CV is on the form.
The code is the credentials.

What we need to ship.

The core is built.

The cryptography is built and measured; the external audit is still ahead.

The chain is ready.

What stands between Protocol 01 and 100 paying merchants on mainnet:

Audit funding

One full external cycle from OtterSec, Neodyme or Trail of Bits. Quote pending. ~\$80K-150K.

First merchant cohort

5 to 10 Solana-native creators or SaaS to seed the verified registry on mainnet. Friend-of-Solana intros highest-leverage.

Ecosystem partnerships

Helius (RPC), Privy (wallet onboarding), Backpack (distribution). Conversations in flight.

Privacy is a right.
Recurring is the rail.
Solana is the chain.

PROTOCOL-01.DEV • @STYX_PQ
VOLTA TEAM • SLASHY FX • SOLO, FULL-TIME
V1.0.3 • FIGURES REFRESHED 2026-09-14